



Establishing electro-functionalized gravity-driven ceramic membrane filtration (EGDCM) for decentralized treatment of algae-laden brackish water: Comparison of *in-situ* electro-oxidation and *ex-situ* electro-coagulation

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ABSTRACT

Algae-laden brackish water (ABW) has remarkably threatened drinking water safety in warm coastal areas. Although gravity-driven ceramic membrane filtration (GDCM) exhibits high potential in ABW treatment during decentralized water supply, membrane fouling is still a critical problem. Herein, GDCM was skillfully electro-functionalized (EGDCM) by *in-situ* electro-oxidation (ISEO) based on self-fabricated Ti/SnO₂-Sb dimensionally stable anode (DSA) (EO-EGDCM) and *ex-situ* electro-coagulation (ESEC) based on iron anode (EC-EGDCM) in this study. EO-EGDCM with KMnO₄ augment (MnEO-EGDCM) was also established for comparison. Results show that ISEO increased GDCM membrane permeability by 22 %, while EC-EGDCM membrane flux was nearly 4.8 times that of GDCM. ISEO enhanced the early removal of organic pollution, and KMnO₄ facilitated the active chlorine oxidization of ammonia and algal toxins on electrified Ti/SnO₂-Sb DSA by suppressing the transformation of free chlorine to less reactive chloramines. Both algae cell permeabilization and intracellular organic matter release were enhanced by ISEO. But SEM-EDS, CLSM and biomass analysis evidenced that membrane biological process, which was improved by 32 %~323 % by electrical stimulation, developed porous structures in the fouling layer in EO-EGDCM/MnEO-EGDCM. According to energy consumption and carbon emissions evaluation, GDCM was confirmed as an energy-saving system for treating ABW with the consumption of only 3.47×10^{-3} kWh/m³. Electricity demand was increased for EGDCM but still considerably lower than that for other algae-laden water treatment processes. EC-EGDCM reduced energy consumption and carbon emission by around 80 % compared to EO-EGDCM/MnEO-EGDCM. Electro-functionalization was a promising option to improve GDCM treatment of ABW via multiple mechanisms but further optimization was still required.

Abbreviations

Abbreviation	Full name
ABW	Algae-laden brackish water
AOM	Algae organic matter
ATP	Adenosine triphosphate
CLSM	Confocal laser scanning microscope
COD	Chemical oxygen demand
CV	Cyclic voltammetry
DSA	Dimensionally stable anode

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DFT	Density functional theory
EASA	Electrochemical active surface area
EDS	Energy dispersive X-ray spectroscopy
EEM	Excitation-emission-matrix
EC-EGDCM	Electro-functionalized gravity-driven ceramic membrane filtration with <i>ex-situ</i> electro-coagulation
EO-EGDCM	Electro-functionalized gravity-driven ceramic membrane filtration with <i>in-situ</i> electro-oxidation
EGDCM	Electro-functionalized gravity-driven ceramic membrane filtration

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